# Can We "Explain" the Decline in Fertility?

A thinking exercise built on trends in the TFR (total fertility rate)

Part 1: Observe → Infer → Examine assumptions → Examine explanations

# Stages of Thinking

## 1 Observe

Survey the graph and put what you see into words

## 2 Infer

"If the observed trend continues…"

"If A…" "If B…"

## 3 Question assumptions & conclusions

What is the conclusion?

On what grounds did you assume that "if"?

What is the conclusion?

## 4 Question the explanation

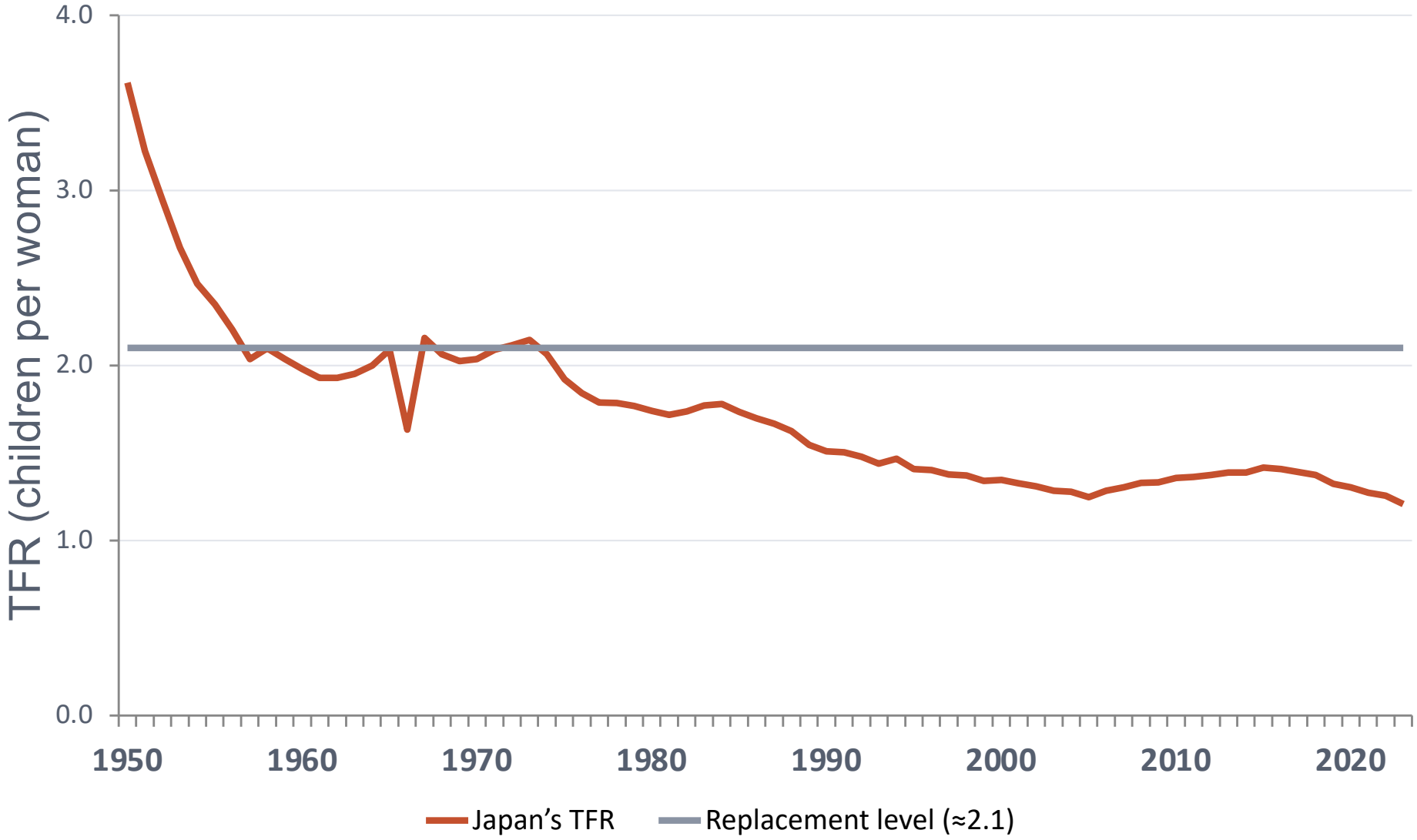
Does the explanation really explain anything?

# 01

# Look at Japan's TFR

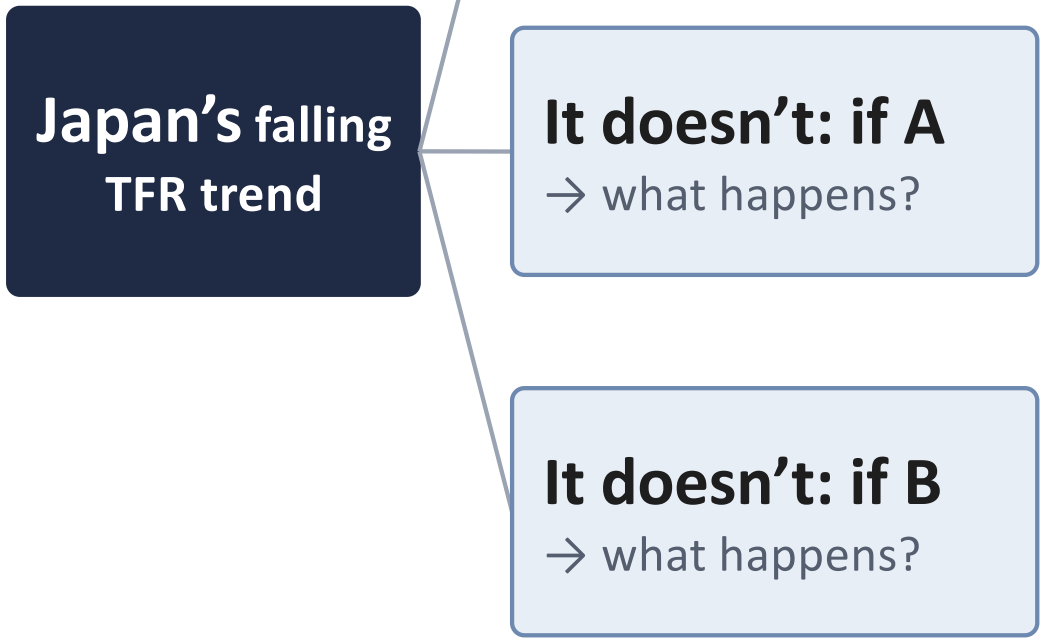
First, from one country's change, build a natural inference

# Japan’s TFR fell from 3.6 to 1.2 in just over 70 years




Source: UN World Population Prospects 2024 (R package wpp2024).

# “If this trend continues…” — a fork in the reasoning




? What reasoning lies behind each “if” —

“if it continues”, “if A”, “if B”?

How should we treat an “if” whose grounds cannot be stated?

# In scientific method, “if it continues” is the natural starting point

## “If it continues” is the baseline

- Extend the observed trend until another reason is shown
- Add no extra assumptions (turning points, causes of reversal)
- Whoever denies it bears the burden of proof

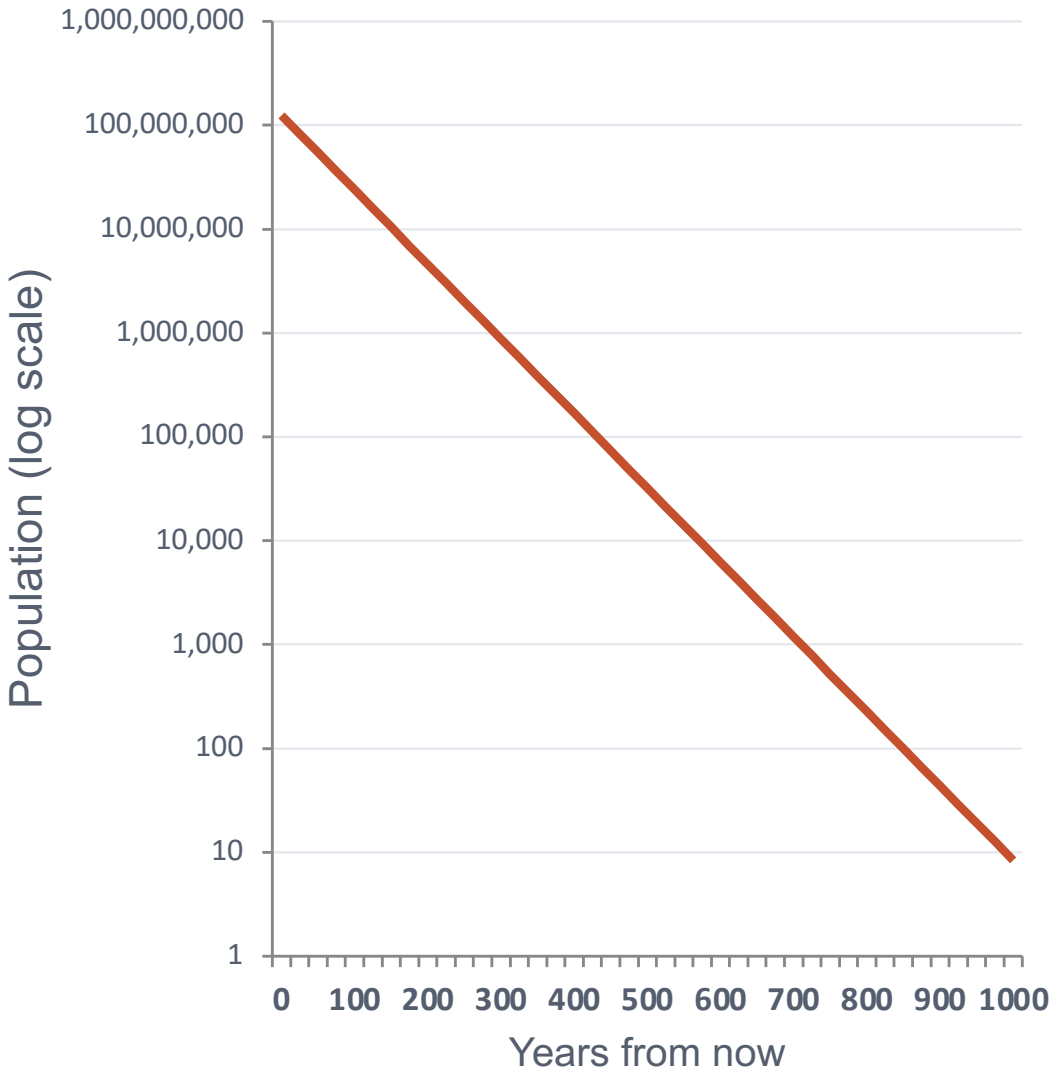
## The role of simple models

- The law of falling bodies is not refuted by “but in reality there is air resistance…”
- A simple model exists not to be refuted, but to make clear what it would take to refute it

?

When we treat “if it continues” as natural, which methodological principle are we adopting? Try to name it.

# If it continues: “extinction” as the outcome




**λ ≈ 0.983622**

**This population carries “extinction” within it.**

**This is the most optimistic case**

If the TFR decline itself continues, the end comes sooner.

| | |
|---|---|
| After 100 years | **≈ 24 million** |
| After 500 years | **≈ 32,000** |
| After 1,000 years | **≈ 8 people** |

Start: Japan’s 2023 population (≈124 million, WPP 2024). λ is given. A simple exponential calculation, not a population projection.

# To reject this conclusion: three conditions for an objection

**1**

**Does it change the direction?**

Does it turn $\lambda < 1$ into $\lambda \geq 1$? Changing only speed or timing just revises the same outcome

**2**

**Can you state the mechanism?**

Not just “we should consider X”, but “considering X makes $\lambda \geq 1$, because…”

**3**

**Is there evidence?**

Is that mechanism shown by data as solid as the original trend?

**?**

Write down your “but in reality…” and test it against the three conditions. Which survive?

If you try to build an objection that meets all three, what question do you run into?

# 02

# What is happening in other countries?

Open tfr_app.html and survey 236 countries and areas, 1950–2023

# First, with your own eyes: survey the world with tfr_app.html

## How to observe

- Double-click tfr_app.html to open it in a browser
- Use the year slider and "▶️ Play" to run from 1950 to 2023
- Toggle countries on and off; compare Japan, China, the US…
- Note differences by continent, and in when and how fast the decline begins

?

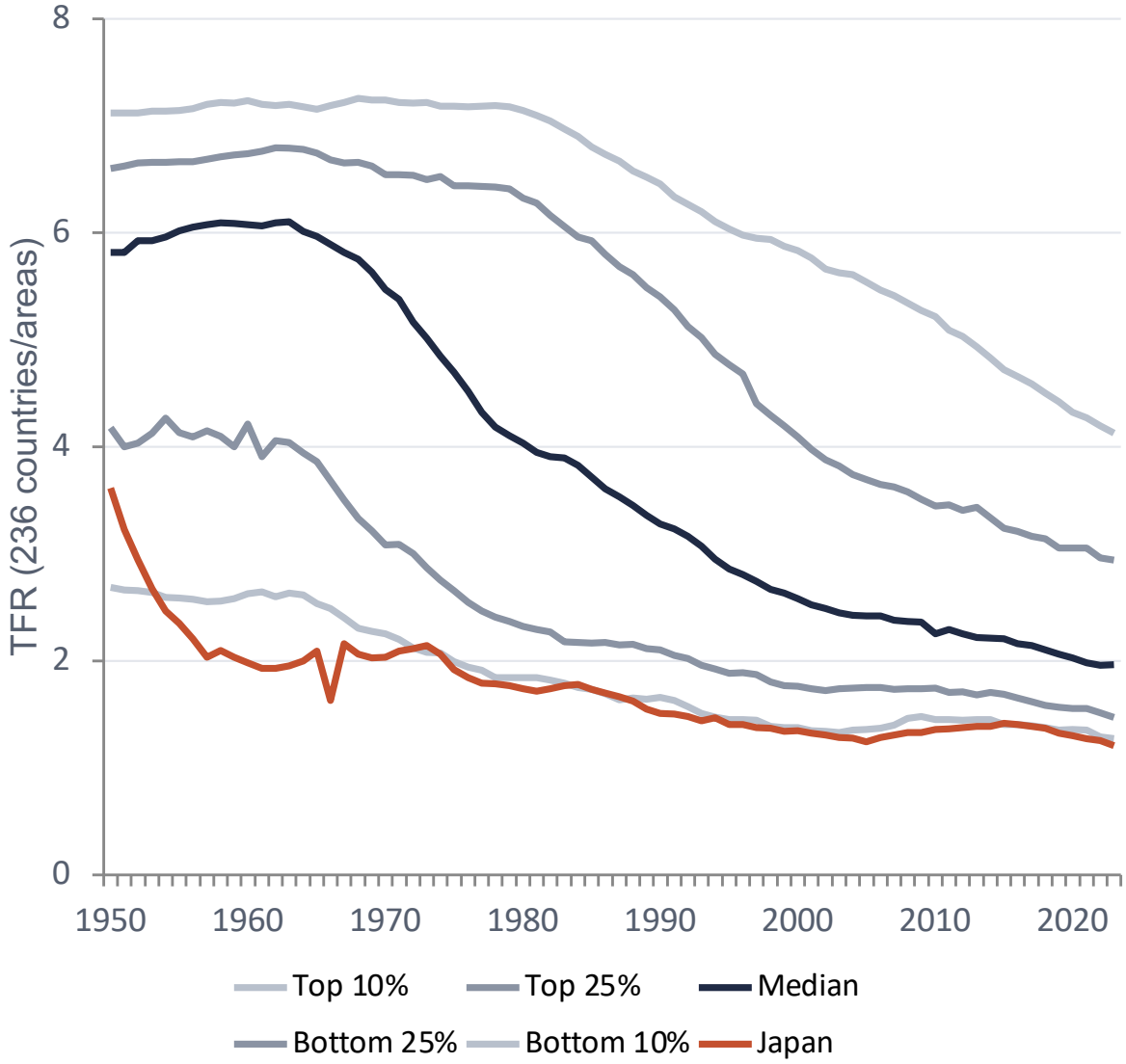
Having surveyed it, what do you see?

What broad trend is visible?

Are there exceptions?

Are they big enough to change the overall picture?

# Almost without exception, countries' TFRs are falling




**233 / 236**

countries/areas with a lower TFR in 2023 than in 1950

**5.82 → 1.96**

median TFR across countries (1950 → 2023)

**1% → 55%**

share of countries/areas with TFR below 2.1

Source: UN WPP 2024 (wpp2024); quantiles of 236 countries/areas each year. The 3 exceptions: Central African Republic, Chad, DR Congo (all still high).

# Does the first feature lead to a “surprising conclusion”?

## The argument used for Japan

- If the trend continues → the population shrinks exponentially
- About 8 people after 1,000 years

## Apply the same argument to the world

- World TFRs are also falling almost without exception
- If this trend continues → ?

? What “surprising conclusion” do you think I was hinting at?

Or is “surprising” wrong — is there nothing to be surprised about?

?

# “Humankind will eventually disappear.”

If we accept the same argument as for Japan, we must accept this inference and its conclusion too.

Wouldn’t biologists and laypeople alike find this “surprising”?

# Is disappear Not surprising — or surprising?

### If it is NOT surprising

- Because the conclusion is “obvious”?
- At what level is it “obvious”?
- Is that level the only layer of thought?
- Shouldn’t the conclusion be examined more deeply at other layers?

### If it IS surprising

- Then the inference itself may be wrong
- What other inferences are there? One, or many?
- How should their validity be measured?
- What conclusion does each lead to?
- Do the “but in reality…” points from Japan hold for the world?

# 03
# Why is this happening?

The TFR decline is common worldwide — a highly general, perhaps universal, phenomenon

# Four commonly cited explanations

**1 Lower infant mortality**

No longer a need to “have many children” to secure labor or heirs

**2 Women’s higher education & employment**

Careers and childbearing collide; later marriage and later births everywhere

**3 Rising cost of raising children**

A shift to “quality”: heavy investment in a few children naturally limits their number

**4 Urbanization & lifestyle change**

Children who were “labor” on farms become “consumers” who raise living costs in cities

# Can we say we “understand” with these explanations?

**For those who stop at surface thinking**

These explanations are the end point of understanding.

**For those who reject surface explanations**

These explanations are a mass of inadequacies.

?

Which side are you on?

For each of the four explanations, name one thing inside it that is still unexplained.

# Questioning Explanation 1: is the “heir” the same in every family?

| Family type (E. Todd) | Example regions | Inheritance |
|---|---|---|
| Absolute nuclear | England, US | Free bequest by parents; weak idea of a designated heir |
| Egalitarian nuclear | N. France, Spain | Equal division among siblings |
| Stem | Japan, Germany, Korea | One heir inherits the house; the “heir” is central |
| Exogamous community | Russia, China | Brothers live with parents and inherit equally |

?

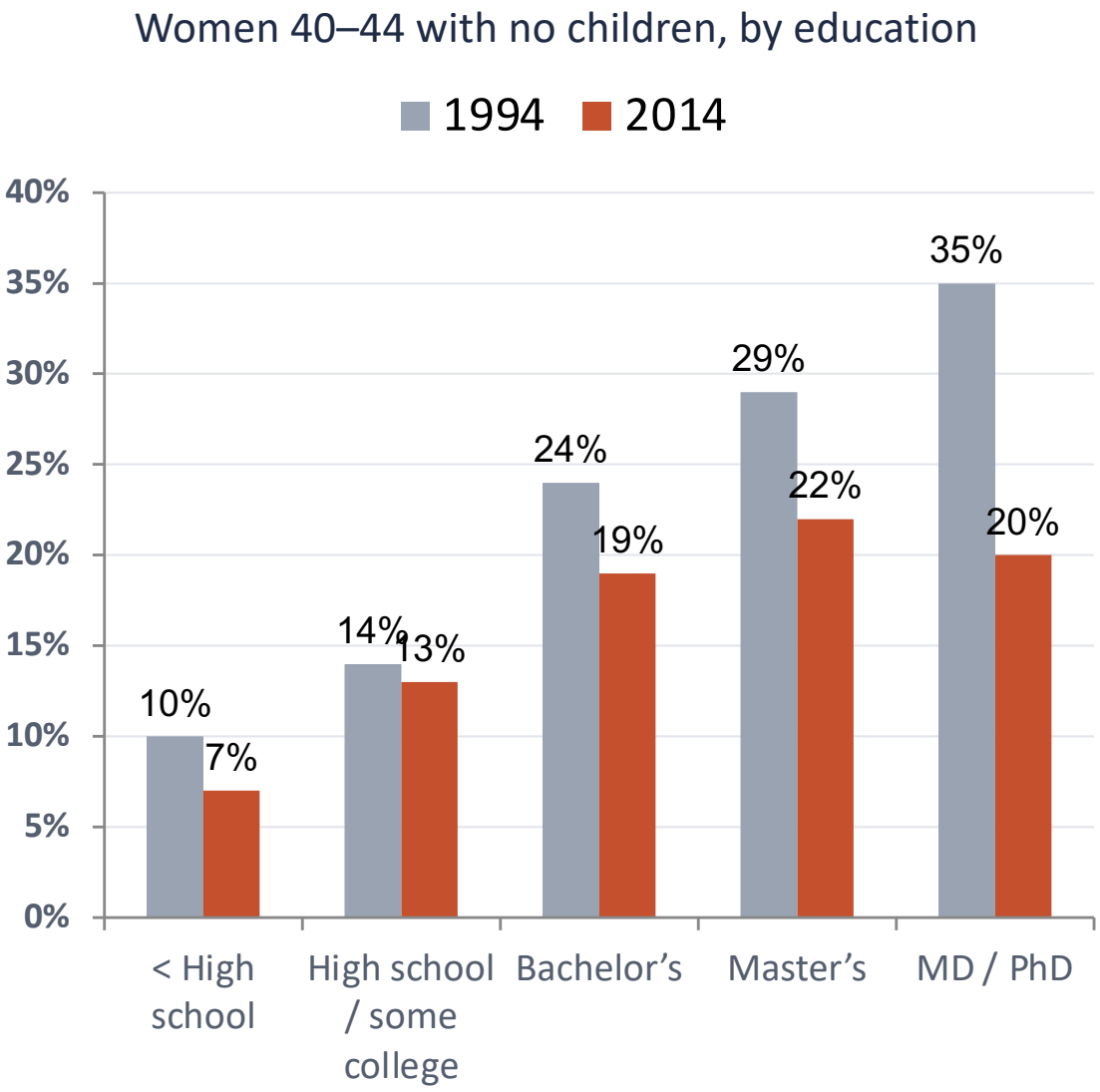
One explanation: “infant mortality fell, so worry about heirs eased.”

But is it acceptable to assume — insensitively — that the “heir” works as one uniform concept across all of Todd’s family systems?

→ Ask an AI this question too.

Family types after E. Todd, La diversité du monde. Regional examples are simplified.

# Questioning Explanation 2: education and children in the US




**Mean children per mother (2014)**

< High school ≈ 2.9
HS / some college ≈ 2.4
Bachelor's or more ≈ 2.2

?

"More education, fewer children" — is this a one-way, fixed relationship?

What does the 20-year fall in childlessness among highly educated women say to Explanation 2?

Source: Pew Research Center (2015), "Childlessness Falls, Family Size Grows Among Highly Educated Women" (US CPS 1992–95, 2012–14).

# Questioning Explanation 3: what is “the cost of raising children”?

**Surface answer**

Education and childrearing costs rose, so people have fewer children

**What theory or logic makes that answer seem acceptable?**

**Should arguments built on that theory be accepted uncritically?**

? 

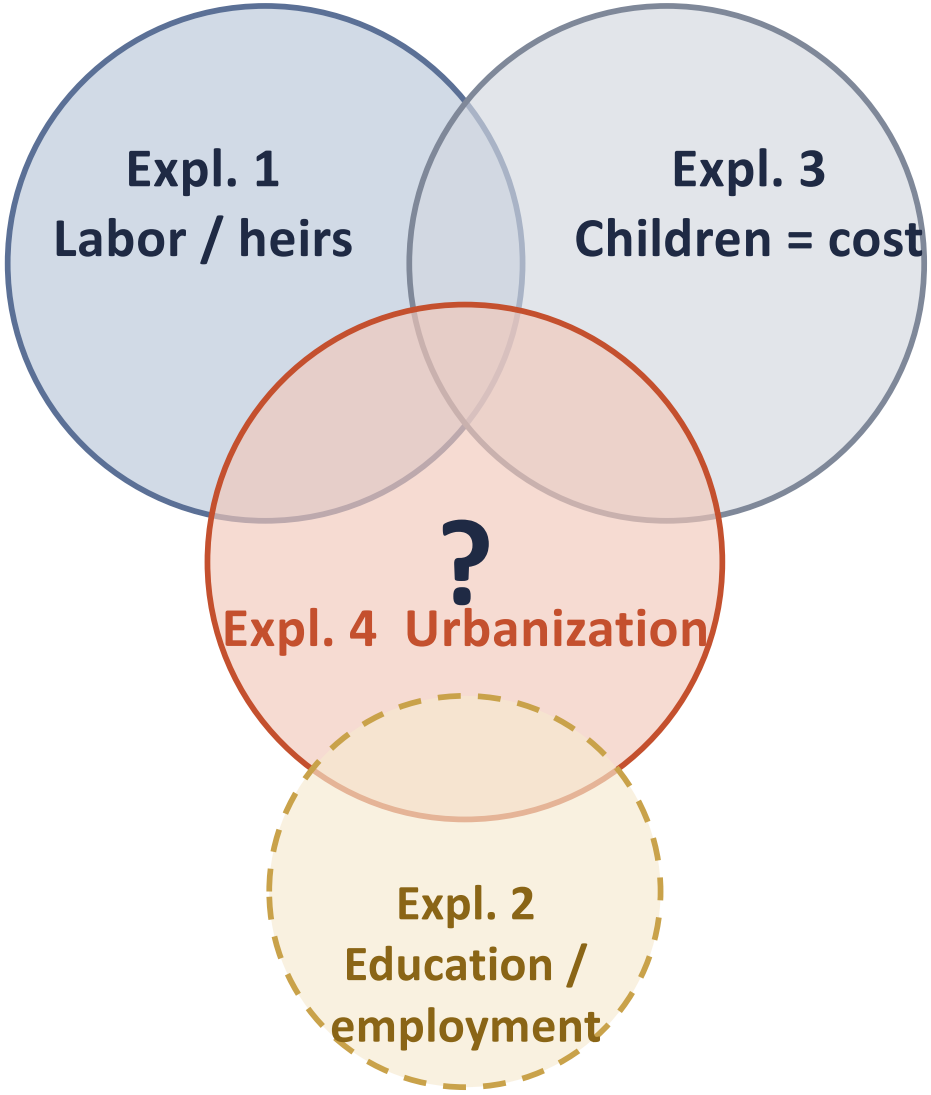
“Cost” — to whom, compared with what?

Where does the idea of children as a “costly good” come from?

Applying that idea, what becomes invisible?

↓ *The deeper you go, the more you see the premises the explanation rests on*

# Questioning Explanation 4: what is unique to urbanization?




?

Explanation 4 overlaps the logic inside Explanations 1, 2 and 3.

Then what element belongs only to this explanation?

Doesn't that element need to be examined?

The overlaps of the dashed circle (Expl. 2) with Expl. 1 and 3 are not yet examined.

# It seems these explanations explain almost nothing.

So, what should we do?

**This is not a conclusion. It is the starting point of a long journey of thought.**

# Materials and sources

- United Nations, DESA, Population Division (2024). World Population Prospects 2024. R package wpp2024 v1.1-3.
- Teaching app: tfr_app.html (Materials / data_graph_interactive_v2 / ja / for_students)
- Livingston, G. (2015). Childlessness Falls, Family Size Grows Among Highly Educated Women. Pew Research Center.
- Todd, E. La diversité du monde: Famille et modernité. (Japanese tr. Ogino F., Fujiwara Shoten)